

LESSON 9.5

ROTATIONS ON THE COORDINATE PLANE



LESSON INTRODUCTION

MA.8.GR.2.3 Describe and apply the effect of a single transformation on two-dimensional figures using coordinates and the coordinate plane.

LEARNING TARGETS

- I can perform a rotation on the coordinate plane.
- I can describe relationships between figures after a rotation.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.912.GR.2.1 Given a preimage and image, describe the transformation and represent the transformation algebraically using coordinates.

MA.912.GR.2.5 Given a geometric figure and a sequence of transformations, draw the transformed figure on a coordinate plane.

ESSENTIAL QUESTIONS

- How do you perform a rotation on the coordinate plane?
- What information do you need to perform a rotation?

LESSON NARRATIVE

Students discover how to use the angle, direction, and center of rotation to perform rotations around the origin on the coordinate plane. Students also describe the relationship between the preimage and image.

COMPONENT SUMMARY

9.5.1 WARM-UP (~5 MIN)

Describe real-world examples of transformations

9.5.3 GUIDED INSTRUCTION (20 MIN)

Discover how to perform rotations on the coordinate plane

9.5.5 GUIDED PRACTICE (5 MIN)

Practice performing rotations on the coordinate plane

9.5.2 EXPLORATION (10 MIN)

Determine degrees of rotations

9.5.4 COLLABORATION (5 MIN)

Discuss which quadrant the image will be in

9.5.6 WRAP-UP (~5 MIN)

Reflect on work

RESOURCES

GLOSSARY

Key Terms: N/A

Additional Terms:

- Rotation
- Mapping
- Clockwise
- Counterclockwise

MATERIALS

- Protractor
- Straightedge
- (Optional) Graph Paper
- (Optional) Patty Paper

ADDITIONAL PREPARATION

- Enlarged visual representations of figures could be created on card stock and cut out prior to lesson.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may interchange clockwise and counterclockwise rotations.
- Students may not understand how the angle of rotation dictates the quadrant that the image is in.

THINGS TO CONSIDER

- Consider reviewing the way analog clock hands move and providing a visual of a clock to help students understand the terms clockwise and counterclockwise.
- Consider reviewing the numbering of the quadrants and where a single point will be after a 90° , 180° , 270° , and 360° rotation. Encourage students to rotate figures one point at a time.
- If time is an issue, then consider assigning Guided Practice for homework.

LITERACY AND LANGUAGE ROUTINES

Think-Pair-Share

In 9.5.2 Exploration, prompt students initially to attempt the questions independently. Then, ask them to discuss how they obtained their answers with a partner. Finally, debrief by having a variety of students share their solutions. These discussions will help build student confidence and understanding of determining angle of rotation.

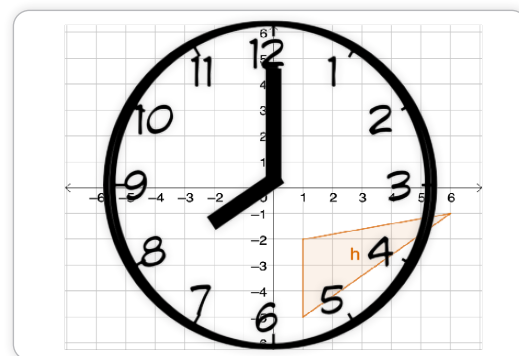
MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.4.1 Discussion and Reflection

In 9.5.2 Exploration and in 9.5.4 Collaboration, students communicate mathematical ideas. In 9.5.2 Exploration, students revisit how the degree of rotation can be found by mapping the vertices thorough the center of rotation. By revisiting the concept, students are given the opportunity to recall what they have learned and to verbalize it to their peers. In 9.5.4 Collaboration, discuss with their partner how to determine which quadrant the image will be in after a rotation about the origin of the preimage.

DIFFERENTIATE INSTRUCTION

Consider providing an auditory entry-point by allowing students to discuss the rotations with a partner in 9.5.5 Guided Practice. Consider providing a visual entry-point by prompting students to initially draw the clock features on coordinate plane to help distinguish between clockwise and counterclockwise directions in 9.5.3 Guided Instruction. Using a protractor and straightedge to measure the angles that connect the vertices and the centers of rotation provides a kinesthetic entry-point in 9.5.2 Exploration.



9.5.1 WARM-UP: GEOMETRY IN ART

TEACHER MOVES

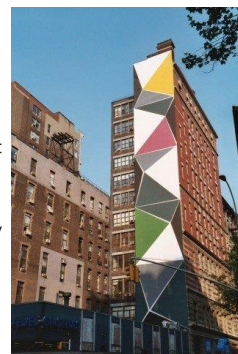
While sequences of transformations are not done in this course, students can reason about this without using that language. It is not intended that all students will recognize a rotation and a translation. Rather, the point is to have them consider a provocative question about how we can see geometry in art.



9.5.1 Warm-Up: Geometry in Art

Tania was a Jewish American artist who was born in Poland and based in New York City. She was an abstract painter, sculptor, and collage artist. She was famous for painting wall art throughout the city.

One of her most well-known pieces of wall art was a 13-story geometric wall painting she created in 1970 that still stands today. It is shown in the picture to the right.



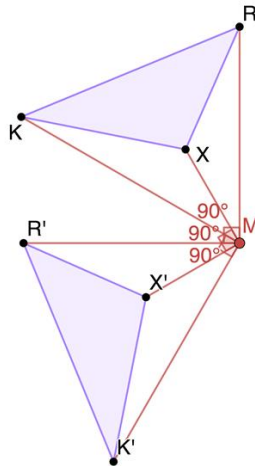
1. Consider the yellow and green triangles shown in the artwork.
 - a. Explain whether you think a single transformation could map the yellow triangle to the green triangle.
Answers will vary. I think it's possible that the transformation could be a single rotation of 180°, if the center of rotation is in the right place.
 - b. Which two transformations do you think could map the yellow triangle to the green triangle if one was performed followed by the other?
Answers will vary. I think you could translate the yellow triangle down and then rotate it to map it onto the green triangle.



9.5.2 Exploration: Determining the Degree of Rotation

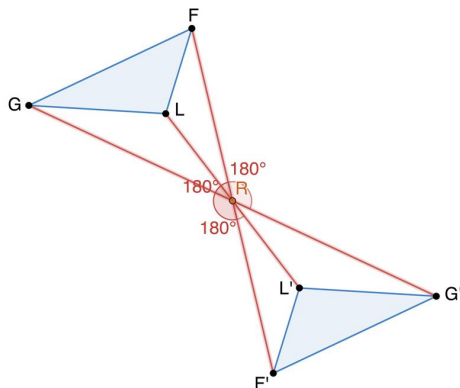
Work with your partner to complete the following.

- The diagram shows $\triangle RKX$ and $\triangle R'K'X'$ after a rotation about point M .



- Use a straightedge to draw $\angle XMX'$, $\angle RMR'$, and $\angle KMK'$.
- Discuss with your partner how the degree of rotation can be determined using the angles that connect the vertices and the center of rotation.

- The diagram shows $\triangle GFL$ and $\triangle G'F'L'$ after a rotation about point R . Use a straightedge to draw $\angle GRG'$, $\angle FRF'$, and $\angle LRL'$.



- Work with your partner to complete each statement using the angles formed by the corresponding vertices of each preimage and image and the center of rotation.
 - Triangle $R'K'X'$ is the result of rotating $\triangle RKX$ 90° counterclockwise about point M .
 - Triangle GFL was rotated 180° counterclockwise about point R to create $\triangle G'F'L'$.

9.5.2 EXPLORATION: DETERMINING THE DEGREE OF ROTATION

ENRICHMENT

- What do you notice about the angle measurements?
- If your measurements are different, are they approximately equal?
- What classroom objects could you use to check if the angle measurements are the same?

TEACHER MOVES

Think-Pair-Share

Prompt students initially to attempt the questions independently. Then, ask them to discuss how they obtained their answers with a partner. Finally, debrief by having a variety of students share their solutions. These discussions will help build student confidence and understanding of determining the angle of rotations.

DIFFERENTIATE INSTRUCTION

Using a protractor and straightedge to measure the angles that connect the vertices and the centers of rotation provides a kinesthetic entry-point.

ENRICHMENT

- Is this rotation clockwise or counterclockwise?
- Why are both answers correct?
- Did the direction of the rotation matter when the angle of rotation was 90°?

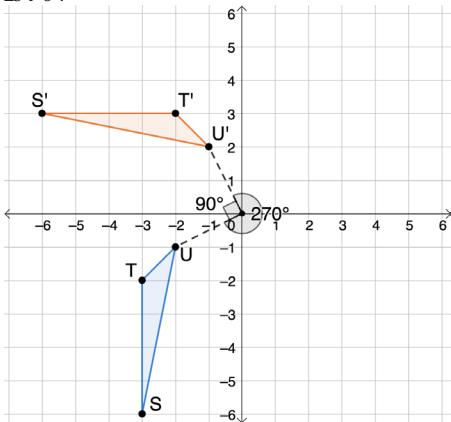
9.5.3 GUIDED INSTRUCTION: ROTATIONS ON THE COORDINATE PLANE

TEACHER MOVES

Prompt students to rotate one point at a time. Remind students to label the vertices of the image with prime notation. After rotating each vertex, ask students to connect the vertices to form the figure of the image.

9.5.3 Guided Instruction: Rotations on the Coordinate Plane

1. On the coordinate grid below, $\triangle STU$ is rotated to create $\triangle S'T'U'$.



- a. There are two ways to describe the rotation. Complete both statements using the information given in the figure.

$\triangle STU$ was rotated 90° clockwise about the origin to create $\triangle S'T'U'$.

$\triangle STU$ was rotated 270° counterclockwise about the origin to create $\triangle S'T'U'$.

ENRICHMENT

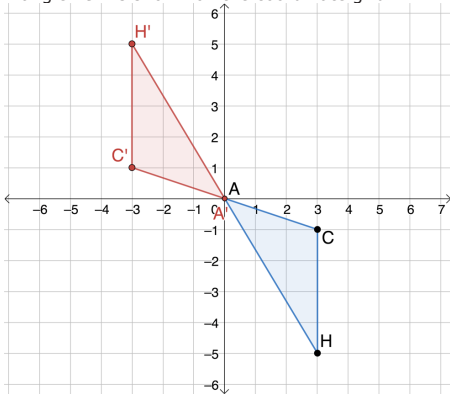
- What is the angle measurement of a full rotation in a circle?
- What do you notice about the sum of the angles of the two equivalent rotations?

- b. For every rotation clockwise, there is a rotation counterclockwise that would create the same image. Complete the table.

Clockwise Rotation	Equivalent Counterclockwise Rotation	Sum of Clockwise & Counterclockwise
90°	270°	360°
180°	180°	360°
270°	90°	360°

- c. Explain what would happen if a figure were rotated 360° with the origin as the center of rotation.
Answers will vary. If a figure was rotated 360° with the origin as the center of rotation, it would end up back in exactly the same place. The image would have the same location and orientation as the preimage.

2. Triangle ACH is shown on the coordinate grid.



Triangle $A'C'H'$ is the image created by rotating $\triangle ACH$ 180° clockwise about the origin.

- a. To show the rotation, follow these steps.
- Because a 180° angle is a straight line, use a straightedge to draw a dotted line from C through the origin that extends to the end of the grid.
 - Plot and label point C' on the dotted line so that $\overline{A'C'}$ is congruent to $\overline{AC}$.
 - Use a straightedge to draw a dotted line from H through the origin that extends to the end of the grid.
 - Plot and label point H' on the dotted line so that $\overline{H'A'}$ is congruent to $\overline{HA}$.
 - Connect the vertices to create $\triangle A'C'H'$.

- b. Describe how point A and point A' relate to each other.

Answers will vary. Point A is in the same place as point A' because it was the center of the rotation. Both are at the origin.

- c. Complete the table with the coordinates of each point.

	Coordinates		Coordinates
Point A	(0, 0)	Point A'	(0, 0)
Point C	(3, -1)	Point C'	(-3, 1)
Point H	(3, -5)	Point H'	(-3, 5)

- d. What do you notice about the coordinates of the corresponding points in the preimage and the image after this rotation?

Answers will vary. I notice that the coordinates of the corresponding points in the preimage and image are opposites of one another.

9.5.3 GUIDED INSTRUCTION: ROTATIONS ON THE COORDINATE PLANE (CONTINUED)

DIFFERENTIATE INSTRUCTION

Prompt students to describe how to perform the rotations to a partner. This will provide an auditory entry-point and give students an opportunity to refine and clarify their thinking.

DIFFERENTIATE INSTRUCTION

It is not intended for students to answer question 2d using an algebraic description.

9.5.4 COLLABORATION: WHICH QUADRANT?

DIFFERENTIATE INSTRUCTION

Consider prompting students to draw a clock on top of the coordinate plane to provide a visual entry-point of the difference between clockwise and counterclockwise.

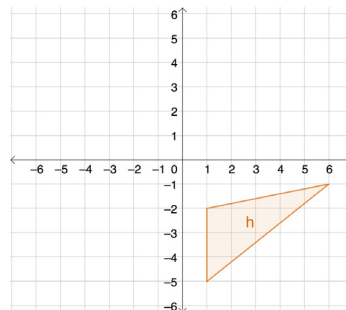
ENRICHMENT

How many quadrants will the figure move if the angle of rotation is 90° , 180° , 270° , 360° ? How is the number of quadrants that the figure moves related to the angle of rotation?



9.5.4 Collaboration: Which Quadrant?

1. A triangle is shown on the coordinate plane.



- a. Without performing the transformation, discuss with your partner what the image of the triangle would look like after a rotation 90° clockwise about the origin.

- b. Complete the statement.

The preimage of the triangle is in quadrant

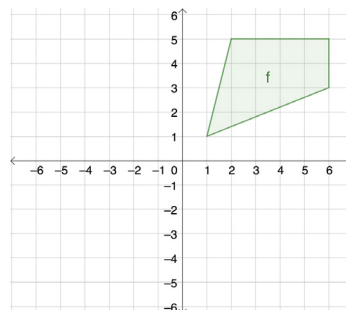
- ☐ 1.
- ☐ 2.
- ☐ 3.
- ☒ 4.

The image of the triangle after a 90° clockwise

rotation about the origin would lie in quadrant

- ☐ 1.
- ☐ 2.
- ☒ 3.
- ☐ 4.

2. A quadrilateral is shown on the coordinate plane.



Without performing the transformation, complete the statement.

The preimage of the quadrilateral is in quadrant

- ☒ 1.
- ☐ 2.
- ☐ 3.
- ☐ 4.

The image of the quadrilateral after a 270°

counterclockwise rotation about the origin would lie in

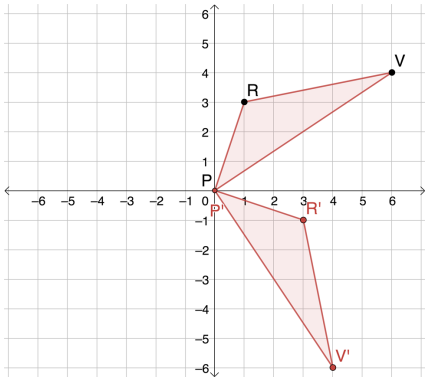
quadrant

- ☐ 1.
- ☐ 2.
- ☐ 3.
- ☒ 4.



9.5.5 Guided Practice: Rotations on the Coordinate Plane

1. Triangle PRV is plotted on the coordinate grid.



- a. Rotate $\triangle PRV$ 270° counterclockwise with the origin as the center of rotation. Label the image $\triangle P'R'V'$.
- b. What is another way to describe the rotation performed in Part A?
Rotation 90° clockwise about the origin.

9.5.5 GUIDED PRACTICE: ROTATIONS ON THE COORDINATE PLANE

QUESTIONING

Consider prompting students who finish early to also rotate the figure 90° and 180° . Consider showing students examples of tessellation patterns and prompt them to create their own as time allows.

ENRICHMENT

Write out the coordinates of the vertices of the preimage and image. What do you notice about the corresponding coordinates?



9.5.6 Wrap-Up: Reflection

1. Take a **SELFIE** by checking the box that best matches what you did today.
Answers will vary.

Showed my work	☐	☐
Explained my answers	☐	☐
Lots of math vocabulary used	☐	☐
Found multiple solutions	☐	☐
I persevered through the problem	☐	☐
Eliminated careless errors	☐	☐

9.5.6 WRAP-UP: REFLECTION

TEACHER MOVES

Consider asking students to include the reasons why they did not do any of the things they mark with a thumbs down. Prompt students to set goals to make more effort in the areas they marked with a thumbs down for this lesson.

DIFFERENTIATE INSTRUCTION

Consider reading the indicators aloud to provide an auditory entry point.